

Multiple Choice

1. **Answer:** A

Options: A) Ne; B) CO₂; C) Ar; D) CH₄; E) O₂

Note: Correct option: A - Ne

2. **Answer:** C

Options: A) In⁺, Sc³⁺ and Se²⁻; B) Ca²⁺, Ga³⁺ and In³⁺; C) Ca²⁺, Sc³⁺ and Se²⁻; D) In³⁺, Ga³⁺ and Se²⁻; E) Ca²⁺, In⁺ and Se²⁻

Note: Correct option: C - Ca²⁺, Sc³⁺ and Se²⁻

3. **Answer:** A

Options: A) CO; B) CO₂; C) CH₄; D) BF₃; E) PCl₅

Note: Correct option: A - CO

4. **Answer:** D

Options: A) 22.1 MeV; B) 23.3 MeV; C) 20.5 MeV; D) 21.2 MeV

Note: Correct option: D - 21.2 MeV

5. **Answer:** A

Options: A) The atomic radius of hydrogen is five times the first Bohr radius; B) The atomic radius of hydrogen is four times the first Bohr radius; C) The atomic radius of hydrogen is half the first Bohr radius; D) The atomic radius of hydrogen is one tenth the first Bohr radius; E) The atomic radius of hydrogen is larger than the first Bohr radius

Note: Correct option: A - The atomic radius of hydrogen is five times the first Bohr radius

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The smell of elements'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1.41 D'.

Long Form Response

11. **Answer:** 1/2 hour. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 50·*Explain why this is the correct answer and provides supporting evidence. Reference key facts.*

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key